

Practical 3

Simplex Method

Use the simplex method to solve the following Lpp

- 1) Maximize $z = 7x_1 + 5x_2$
Subject to linear constraint,
$$x_1 + 2x_2 \leq 6$$
$$4x_1 + 3x_2 \leq 12$$

And
$$x_1 \geq 0, x_2 \geq 0$$

- 2) Maximize $z = 3x_1 + 5x_2 + 4x_3$
Subject to linear constraint,
$$x_1 + 2x_2 + x_3 \leq 5$$
$$2x_1 + x_2 + 3x_3 \leq 8$$
$$x_1 + x_2 + 2x_3 \leq 6$$

And
$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$